

horizon-fixed evolution scheme using BS coordinates

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1 vacuum case

Lagrangian density

$$\mathcal{L} = R - 2\Lambda \quad (1.1)$$

Einstein equation

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 0 \quad (1.2)$$

Calculate the trace on both sides of the equation

$$R - 2R + 4\Lambda = 0 \Rightarrow R = 4\Lambda \quad (1.3)$$

So the Einstein equation is equivalent to

$$R_{\mu\nu} - \Lambda g_{\mu\nu} = 0 \quad (1.4)$$

2 with scalar field

Lagrangian density

$$\mathcal{L} = R - \frac{1}{2}\nabla_\mu\phi\nabla^\mu\phi - P(\phi) \quad (2.1)$$

where

$$P(\phi) = -6\cosh\left(\frac{\phi}{\sqrt{3}}\right) + b_4\phi^4 \quad (2.2)$$

and $b_4 = -\frac{1}{5}$.

the BS coordinates

$$ds^2 = -\left(e^{2\beta}Vz - \frac{h_{AB}U^AU^B}{z^2}\right)dt^2 - \frac{2e^{2\beta}}{z^2}dt\,dz - \frac{2h_{AB}U^B}{z^2}dt\,dx^A + \frac{h_{AB}}{z^2}dx^A\,dx^B \quad (2.3)$$

where

$$h = e^{2\chi} \begin{pmatrix} e^\alpha \cosh \theta & \sinh \theta \\ \sinh \theta & e^{-\alpha} \cosh \theta \end{pmatrix}, \quad (U^A) = \begin{pmatrix} \xi \\ \eta \end{pmatrix} \quad (2.4)$$

we fix the radial dependence of χ through a gauge choice:

$$\chi(t, x, y, z) = \frac{1}{4} \ln(1 + 2z^3 \chi_3(t, x, y)) \quad (2.5)$$

so

$$\chi' = \frac{3z^2 \chi_3}{2 \cdot (2z^3 \chi_3 + 1)} \quad (2.6)$$

$$\chi_x = \frac{z^3 \chi_{3x}}{2 \cdot (2z^3 \chi_3 + 1)} \quad (2.7)$$

$$\chi'' = -\frac{9z^4 \chi_3^2}{(2z^3 \chi_3 + 1)^2} + \frac{3z \chi_3}{2z^3 \chi_3 + 1} \quad (2.8)$$

$$\chi'_x = -\frac{3z^5 \chi_3 \chi_{3x}}{(2z^3 \chi_3 + 1)^2} + \frac{3z^2 \chi_{3x}}{2 \cdot (2z^3 \chi_3 + 1)} \quad (2.9)$$

$$\chi_{xx} = -\frac{z^6 (\chi_{3x})^2}{(2z^3 \chi_3 + 1)^2} + \frac{z^3 \chi_{3xx}}{2 \cdot (2z^3 \chi_3 + 1)} \quad (2.10)$$

apparent horizon condition:

$$z^2 V(z\chi' - 1) - 2\xi \chi_x - \xi_x - 2\dot{\chi}|_{z=1} = 0 \quad (2.11)$$

i.e.

$$\dot{\chi} - \frac{z^2}{2} V(z\chi' - 1) + \xi \chi_x + \frac{\xi_x}{2}|_{z=1} = 0 \quad (2.12)$$

E_{zz} gives rise to the constraint equation of β :

$$(z\chi' - 1)\beta' = \frac{z}{8}(\alpha'^2 + \phi'^2 + 4\chi'^2) + \frac{z}{2}\chi'' \quad (2.13)$$

E_{zx} gives rise to the constraint equation of ξ :

$$\frac{z^2 e^{-2\chi}}{2} \Xi' = \frac{e^{2\chi}}{z^2} (z^2 e^{-2\chi} \beta_x)' - \frac{1}{2}(\alpha'_x + 2\alpha' \chi_x - \alpha' \alpha_x - \phi' \phi_x) + \chi'_x \quad (2.14)$$

where

$$\Xi = \frac{e^{4\chi - 2\beta + \alpha} \xi'}{z^2} \quad (2.15)$$

$E_{zv} - \frac{f}{2} E_{zz} + \xi E_{zx} + \eta E_{zy}$ gives rise to the constraint equation of $d_t \chi$:

$$\begin{aligned} 2e^{-2\chi} z^2 \left(\frac{e^{2\chi}}{z^2} d_t \chi \right)' &= \frac{e^{\alpha - 2\beta + 2\chi} (\xi')^2}{4} + 2\xi_x \left(\frac{1}{z} - \chi' \right) - \frac{\xi'_x}{2} + \chi_x \left(\frac{4\xi}{z} - 4\xi \chi' - \xi' \right) - 2\xi \chi'_x + \frac{Pe^{2\beta}}{2z^2} \\ &+ e^{-\alpha + 2\beta - 2\chi} \left((\beta_x)^2 + \frac{(\phi_x)^2}{4} - \alpha_x \left(\beta_x + \chi_x - \frac{\alpha_x}{2} \right) + \beta_{xx} + \chi_{xx} - \frac{\alpha_{xx}}{2} \right) \end{aligned} \quad (2.16)$$

$e^{-\alpha} E_{xx} - e^{\alpha} E_{yy}$ gives rise to the constraint equation of $d_t \alpha$:

$$\begin{aligned} e^{-\chi} z \left(\frac{e^{\chi}}{z} d_t \alpha \right)' + \alpha' d_t \chi &= \frac{e^{\alpha - 2\beta + 2\chi} (\xi')^2}{4} + \xi_x \left(\frac{1}{z} - \chi' - \frac{\alpha'}{2} \right) - \frac{\xi'_x}{2} + \alpha_x \left(\frac{\xi}{z} - \xi \chi' - \frac{\xi'}{2} \right) - \xi \chi_x \alpha' - \xi \alpha'_x \\ &+ e^{-\alpha + 2\beta - 2\chi} \left((\beta_x)^2 + \frac{(\phi_x)^2}{4} - 2\beta_x \chi_x + \beta_{xx} \right) \end{aligned} \quad (2.17)$$

the scalar equation gives rise to the constraint equation of $d_t\phi$:

$$e^{-\chi}z \left(\frac{e^\chi}{z} d_t\phi \right)' + \phi' d_t\chi = -\xi_x \frac{\phi'}{2} + \phi_x \left(\frac{\xi}{z} - \xi\chi' - \frac{\xi'}{2} \right) - \xi\chi_x\phi' - \xi\phi'_x - \frac{e^{2\beta} \frac{\partial}{\partial\phi} P}{2z^2} \\ + e^{-\alpha+2\beta-2\chi} \left(\phi_x \left(\beta_x - \frac{\alpha_x}{2} \right) + \frac{\phi_{xx}}{2} \right) \quad (2.18)$$

where

$$d_t\chi = \dot{\chi} - \frac{z^2}{2} V(z\chi' - 1) \quad (2.19)$$

$$d_t\alpha = \dot{\alpha} - \frac{z^3}{2} V\alpha' \quad (2.20)$$

$$d_t\phi = \dot{\phi} - \frac{z^3}{2} V\phi' \quad (2.21)$$

$$(2.22)$$

$E_t^z + U^A E_A^z|_{z=1}$ gives rise to the elliptic equation of V_H :

$$\frac{z^3 e^{-\alpha+2\beta-2\chi}}{2} V_{xx} + z^3 \left(\frac{\xi'}{2} + e^{-\alpha+2\beta-2\chi} \left(\beta_x - \frac{\alpha_x}{2} \right) \right) V_x \\ = \xi^2 \left(\frac{(\alpha_x)^2}{2} + \frac{(\phi_x)^2}{2} \right) + \xi (d_t\alpha\alpha_x + d_t\phi\phi_x) + \xi_x \left(\frac{\xi_x}{2} + \xi\alpha_x + d_t\alpha \right) + \frac{d_t\alpha^2}{2} + \frac{d_t\phi^2}{2} \quad (2.23)$$

or

$$\frac{z^3 e^{-\alpha+2\beta-2\chi}}{2} V_{xx} + z^3 \left(\frac{\xi'}{2} + e^{-\alpha+2\beta-2\chi} \left(\beta_x - \frac{\alpha_x}{2} \right) \right) V_x - z^3 \left(\xi\chi'_x + \chi_x\xi' + \partial_z d_t\chi + \frac{\xi'_x}{2} \right) V \\ = \xi^2 \left(\frac{(\alpha_x)^2}{2} + \frac{(\phi_x)^2}{2} \right) + \xi (d_t\alpha\alpha_x + d_t\phi\phi_x) + \xi_x \left(\frac{\xi_x}{2} + \xi\alpha_x + d_t\alpha \right) + \frac{d_t\alpha^2}{2} + \frac{d_t\phi^2}{2} \\ - 2\xi \left(\xi\chi_{xx} + \xi_x\chi_x + \frac{\xi_{xx}}{2} + \partial_x d_t\chi \right) \quad (2.24)$$

2.1 $\beta_0 = 0$

let $\beta_0 = 0, \xi_0 = 0, \alpha_0 = 0$, asymptotic behavior (near $z = 0$):

$$\beta = -\frac{\phi_1^2}{16}z^2 - \left(\frac{\phi_1\phi_2}{6} + \frac{\chi_3}{2}\right)z^3 + O(z^4) \quad (2.25)$$

$$\Xi = 3\xi_3 + O(z) \quad (2.26)$$

$$\xi = \xi_3 z^3 + O(z^4) \quad (2.27)$$

$$z^3 V = 1 + V_3 z^3 + O(z^4) \quad (2.28)$$

$$\chi = \frac{\chi_3}{2}z^3 + O(z^4) \quad (2.29)$$

$$\alpha = \alpha_3 z^3 + O(z^4) \quad (2.30)$$

$$\phi = \phi_1 z + \phi_2 z^2 + O(z^3) \quad (2.31)$$

$$d_t \chi = \frac{1}{2z} + \frac{\phi_1^2 z}{16} + z^2 \left(\frac{V_3}{2} - \frac{3\chi_3}{4} \right) + O(z^3) \quad (2.32)$$

$$d_t \alpha = -\frac{3z^2 \alpha_3}{2} + O(z^3) \quad (2.33)$$

$$d_t \phi = -\frac{\phi_1}{2} - z\phi_2 + z^2 \left(-\frac{\phi_1^3}{16} - \frac{3\phi_3}{2} + \dot{\phi}_2 \right) + O(z^3) \quad (2.34)$$

equations of $\dot{V}_3$ and $\dot{\xi}_3$:

$$\dot{\xi}_3 = -\frac{2\phi_1 \partial_x \phi_2}{9} + \frac{\partial_x V_3}{3} - \partial_x \alpha_3 - \frac{2\partial_x \chi_3}{3} \quad (2.35)$$

$$\dot{V}_3 = \frac{\phi_1 \dot{\phi}_2}{6} + 2\dot{\chi}_3 + \frac{3\partial_x \xi_3}{2} \quad (2.36)$$

subtraction

$$\beta = -\frac{\phi_1^2}{16}z^2 - \left(\frac{\phi_1\phi_2}{6} + \frac{\chi_3}{2}\right)z^3 + z^4 \beta_s, \quad (2.37)$$

$$\xi = z^2 \xi_s, \quad (2.38)$$

$$\Xi = \Xi_s, \quad (2.39)$$

$$V = \frac{1}{z^3}(1 + z^2 V_s), \quad (2.40)$$

$$d_t \chi = \frac{1}{2z} + \frac{\phi_1^2 z}{16} + z^2 e^{-2\chi} d_t \chi_s, \quad (2.41)$$

$$\alpha = z^2 \alpha_s, \quad (2.42)$$

$$\phi = z \phi_s, \quad (2.43)$$

$$d_t \alpha = z d_t \alpha_s, \quad (2.44)$$

$$d_t \phi = -\frac{\phi_1}{2} + z e^{-\chi} d_t \phi_s, \quad (2.45)$$

$$(2.46)$$

let $d_t\chi = \frac{1}{2z} + \frac{\phi_1^2 z}{16} + z^2 e^{-2\chi} d_t\chi_s$, then

$$\begin{aligned} & 2e^{-2\chi} z^2 \left(\frac{e^{2\chi}}{2z^3} + \frac{e^{2\chi}\phi_1^2}{16z} + d_t\chi_s \right)' \\ &= \frac{e^{\alpha-2\beta+2\chi} (\xi')^2}{4} + 2\xi_x \left(\frac{1}{z} - \chi' \right) - \frac{\xi'_x}{2} + \chi_x \left(\frac{4\xi}{z} - 4\xi\chi' - \xi' \right) - 2\xi\chi'_x + \frac{Pe^{2\beta}}{2z^2} \\ &+ e^{-\alpha+2\beta-2\chi} \left((\beta_x)^2 + \frac{(\phi_x)^2}{4} - \alpha_x \left(\beta_x + \chi_x - \frac{\alpha_x}{2} \right) + \beta_{xx} + \chi_{xx} - \frac{\alpha_{xx}}{2} \right) \end{aligned} \quad (2.47)$$

let $d_t\phi = -\frac{\phi_1}{2} + ze^{-\chi} d_t\phi_s$ then

$$\begin{aligned} e^{-\chi} z \left(-\frac{e^{\chi}\phi_1}{2z} + d_t\phi_s \right)' + \phi' d_t\chi &= -\xi_x \frac{\phi'}{2} + \phi_x \left(\frac{\xi}{z} - \xi\chi' - \frac{\xi'}{2} \right) - \xi\chi_x\phi' - \xi\phi'_x - \frac{e^{2\beta} \frac{\partial}{\partial\phi} P}{2z^2} \\ &+ e^{-\alpha+2\beta-2\chi} \left(\phi_x \left(\beta_x - \frac{\alpha_x}{2} \right) + \frac{\phi_{xx}}{2} \right) \end{aligned} \quad (2.48)$$

2.2 $\phi = 0$ and open source β_0

subtraction

$$\beta = \beta_0 - \frac{z^2}{2}\chi_3 + z^4\beta_4 \quad (2.49)$$

$$(z\chi' - 1)(\beta_0 - \frac{z^2}{2}\chi_3 + z^4\beta_4)' = \frac{z}{8}(\alpha'^2 + \phi'^2 + 4\chi'^2) + \frac{z}{2}\chi'' \quad (2.50)$$

2.3 static solution

equations for static solutions with spherical symmetry:

$$0 = z\beta'\chi' - \frac{z(\chi')^2}{2} - \frac{z\frac{d^2}{dz^2}\chi}{2} - \frac{z(\phi')^2}{8} - \beta' \quad (2.51)$$

$$0 = -2z^3V(\chi')^2 - z^3V\frac{d^2}{dz^2}\chi - z^3V'\chi' + z^2V\chi' + z^2V' - \frac{P(\phi)e^{2\beta}}{2z^2} \quad (2.52)$$

$$0 = z^3Ve^{-2\beta}e^{2\chi}\frac{d^2}{dz^2}\beta + z^3Ve^{-2\beta}e^{2\chi}(\chi')^2 + z^3Ve^{-2\beta}e^{2\chi}\frac{d^2}{dz^2}\chi + \frac{z^3Ve^{-2\beta}e^{2\chi}(\phi')^2}{4} + z^3e^{-2\beta}e^{2\chi}V'\beta' + z^3e^{-2\beta}e^{2\chi}V\beta' \quad (2.53)$$

$$0 = -z^3V\chi'\phi' - \frac{z^3V\frac{d^2}{dz^2}\phi}{2} - \frac{z^3V'\phi'}{2} - \frac{z^2V\phi'}{2} + \frac{e^{2\beta}\frac{d}{d\phi}P(\phi)}{2z^2} \quad (2.54)$$

boundary conditions:

$$\beta|_{z=0} = 0 \quad (2.55)$$

$$\chi'|_{z=1} = 1 \quad (2.56)$$

$$V|_{z=1} = V_0 \quad (2.57)$$

$$\phi'|_{z=0} = \phi_1 \quad (2.58)$$